

Database Programming in Python

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From a construction firm to a stock exchange, every organisation depends on large databases. These are essentially collections of tables, and connected with each other through columns.

These database systems support SQL, the Structured Query Language, which is used to create, access and manipulate the data. SQL is used to access data, and also to create and exploit the relationships between the stored data. Additionally, these databases support database normalisation rules for avoiding redundancy of data.

The Python programming language has powerful features for database programming. Python supports various databases like MySQL, Oracle, Sybase, PostgreSQL, etc. Python also supports Data Definition Language (DDL), Data Manipulation Language (DML) and Data Query Statements.

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Benefits of Python for database programming

There are many good reasons to use Python for programming database applications:

- Programming in Python is arguably more efficient and faster compared to other languages.
- Python is famous for its portability.
- It is platform independent.
- Python supports SQL cursors.
- In many programming languages, the application developer needs to take care of the open and closed connections of the database, to avoid further exceptions and errors. In

Python, these connections are taken care of.

- Python supports relational database systems.
- Python database APIs are compatible with various databases, so it is very easy to migrate and port database application interfaces.

DB-API (SQL-API) for Python

Python DB-API is independent of any database engine, which enables you to write Python scripts to access any database engine. The Python DB API implementation for MySQL is MySQLdb. For PostgreSQL, it supports psycopg, PyGresQL and pyPgSQL modules. DB-API implementations for Oracle are dc_oracle2 and cx_oracle. Pydb2 is the DB-API implementation for DB2. Python's DB-API consists of connection objects, cursor objects, standard exceptions and some other module contents, all of which we will discuss.

Connection objects

Connection objects create a connection with the database and these are further used for different transactions. These connection objects are also used as representatives of the database session.

A connection is created as follows:

```
>>>conn = MySQLdb.connect("library", user="linuxjournal",
password="python")
```

You can use a connection object for calling methods like *commit()*, *rollback()* and *close()* as shown below:

```
>>>cur = conn.cursor() //creates new cursor object for
executing SQL statements
>>>conn.commit() //Commits the transactions
>>>conn.rollback() //Roll back the transactions
```